

Technical Document #703: Natural Language Processing - Comprehensive Overview

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Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Text summarization generates a concise summary of a longer text.
- Tokenization is the process of breaking text into individual units called tokens.
- Coreference resolution identifies all expressions that refer to the same entity in a text.